HCI: Mid Semester Exam: Part A (Take Home Exam)

Weightage in Mid Semester Exam: 50%

Submission Due Date: 16/02/2026

Question 1: Create a video using "Paper Based Design" to demonstrate the HCI design for an imaginary product "Time Machine". Assume that this "Time Machine" can help you time travel to any location on the Earth at any time in past or future. Prepare the video that illustrates the sequences of the screens and the user interaction with the screens. Prepare a report as per the following mentioned guidelines. You must work individually and submit your report individually as per the guidelines. [15 Marks]

Following are the essential details user needs to enter in the UI of the Time Machine. These details should be entered on different screens. You may add any intermediate steps in addition to the following essential details needed for building UI/screens. You may add any additional required screens.

- User name and e-mail.
- To which location on the Earth the user wants to travel.
- To which time (date, month and year) user wants to travel (past or future).
- For how much duration (hours and days) user wants to stay at above location.
- For the return journey, at which location user want to return after the mentioned time travel.

Notes:

(1) For every entry by the user, the system (time machine) should respond appropriately and then move forward to the next screen in the sequence to complete all entries.
(2) You should also prepare at-least two screen to highlight any error by the user as illustrations of good error message.
(3) Use the principles of good HCI design learned in the class for above design.

Guidelines for Preparing the Paper Based Low Fidelity Prototype and the Video:

1. Design all the screens and the user responses to the screen using paper and pen design.
2. You must draw all the screen yourself. You must write all the content. No Printout is allowed.
3. You may use pens of different colors, papers of different colors, sketch pens, or colors of any type.
4. You must imagine and prepare all the screens where the machine seeks the inputs from the user and how user enters the details. You should imagine the machine's response to the user inputs.
5. Prepare a video that starts from the first screen/page of your time machine until the end of the user interaction, documenting all the interactions of the user and the machine in the right sequence.
6. You must use your own voice for the required voice-over in the video.

Guidelines for Preparing the Report:

1. Prepare a report (MS word or equivalent) and describe the details of each page (screen) you have designed with the photo of each page followed by the description.
2. For each screen, describe its Function, Design Principles/Universal Usability Principles utilized/implemented, details of Interaction Style, if applicable, quality of error message if applicable, and any other technical details (HCI principles/guidelines) relevant to that page (screen) in about 25 to 75 words with required annotations and illustrations.
3. On the first page of the report, clearly mentions your name, enrolment number and other relevant details.

Guidelines for Submission:

1. Name your report file as HCI_Mid_Sem_Part_A_2026_< your enrolment number >.
2. Store the video in any online folder/location. Share the location of the video folder by including its link in the above mentioned report on the last page of the report. Give "view" access of that video to me and all TAs.
3. Submit your report individually by replying to this e-mail on or before the end of the day on 16/02/2026. Put all the TAs in the CC of the e-mail.
4. Put all the paper designs you have prepared in one envelop. Write Your Name, Enrolment Number and other required details on that envelop. Submit me that envelop during the lecture on 17/02/2026.

Evaluation Matrix: [Total 15 Marks]

1. Evaluations based on video:
    a. Completeness of the interaction design to achieve all the goals of the design. [2 Mark]
    b. Quality of Paper designs. [3 Marks]
    c. Sound effects, music (if and when required) and Voice Over for user interactions. [2 Mark].
    d. Creativity and Innovative Approaches in the design. [2 Marks]
2. Evaluation based on Report. [6 Marks]
    a. Refer to the guidelines for preparing the report mentioned above.